

**Advanced (Habanero)****Q1.** What is the expansion of  $(x + y)^2$ ?

- (A)  $x^2 + 2xy + y^2$   
 (B)  $x^2 - 2xy + y^2$   
 (C)  $x^2 + y^2$   
 (D)  $2x^2 + 2y^2$   
 (E)  $x^2 - y^2$

**Q2.** Which of the following is a perfect square trinomial?

- (A)  $x^2 + 4x + 4$   
 (B)  $x^2 - 4x - 4$   
 (C)  $x^2 + 2x + 1$   
 (D)  $x^2 - 2x - 1$   
 (E)  $x^2 + 1$

**Q3.** What is the factored form of  $x^2 + 10x + 25$ ?

- (A)  $(x + 5)^2$   
 (B)  $(x - 5)^2$   
 (C)  $(x + 5)(x - 5)$   
 (D)  $(x + 2)(x + 8)$   
 (E)  $(x - 2)(x - 8)$

**Q4.** What is the expansion of  $(x - y)^2$ ?

- (A)  $x^2 + 2xy + y^2$   
 (B)  $x^2 - 2xy + y^2$   
 (C)  $x^2 + y^2$   
 (D)  $2x^2 + 2y^2$   
 (E)  $x^2 - y^2$

**Q5.** Which of the following is not a perfect square trinomial?

- (A)  $x^2 + 6x + 9$   
 (B)  $x^2 - 6x + 9$   
 (C)  $x^2 + 2x + 1$   
 (D)  $x^2 + 2x + 2$   
 (E)  $x^2 - 2x + 1$

**Q6.** What is the factored form of  $x^2 - 12x + 36$ ?

- (A)  $(x + 6)^2$   
 (B)  $(x - 6)^2$   
 (C)  $(x + 6)(x - 6)$   
 (D)  $(x + 2)(x - 8)$   
 (E)  $(x - 2)(x + 8)$

**Q7.** What is the expansion of  $(2x + 3)^2$ ?

- (A)  $4x^2 + 12x + 9$   
 (B)  $4x^2 - 12x + 9$   
 (C)  $4x^2 + 6x + 9$   
 (D)  $4x^2 - 6x + 9$   
 (E)  $4x^2 + 12x - 9$

**Q8.** Which of the following is the factored form of  $x^2 + 8x + 16$ ?

- (A)  $(x + 4)^2$   
 (B)  $(x - 4)^2$   
 (C)  $(x + 4)(x - 4)$   
 (D)  $(x + 2)(x + 6)$   
 (E)  $(x - 2)(x - 6)$

**Open Ended Questions (FRQ)****Q9.** Factor the expression  $x^2 - 14x + 49$ .**Q10.** Find the expansion of  $(3x - 2)^2$  and factor the resulting expression.**Chef's Tips**

- Review the difference between perfect square trinomials and other types of trinomials.
- Make sure students understand how to factor perfect square trinomials.
- Have students practice expanding and factoring various expressions to reinforce their understanding.